C1.1 Enzymes and metabolism (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

9 MARKS

1 **STATE** one example of an intracellular and one of an extracellular enzyme-catalysed reaction. [C1.1.11] [2]

2 **STATE** why metabolic reactions always release some heat. [C1.1.12] [1]

3 **NAME** two cyclical metabolic pathways. [C1.1.13] [2]

4 **NAME** the enzyme that statins inhibit. [C1.1.15] [1]

5 **NAME** the enzyme that isoleucine inhibits. [C1.1.16] [1]

6 **STATE** the enzyme that penicillin inhibits and the effect on the bacterium. [C1.1.17] [2]

Section B: Understanding

15 MARKS

7 **COMPARE** linear and cyclical metabolic pathways. [C1.1.13] [3]

8 **EXPLAIN** how a molecule binding to an allosteric site can inhibit an enzyme. [C1.1.14] [3]

9 **EXPLAIN** how feedback inhibition by isoleucine benefits a bacterial cell. [C1.1.16] [3]

10 **EXPLAIN** how a change in transpeptidase can make bacteria resistant to penicillin. [C1.1.17] [3]

11 **EXPLAIN** how heat from metabolism helps mammals and birds. [C1.1.12] [3]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

12 [C1.1.14, C1.1.15, C1.1.17]

The initial rate of an enzyme reaction was measured at different substrate concentrations, with no inhibitor, with inhibitor P, and with inhibitor Q (same concentration of each inhibitor).

SUBSTRATE / MMOL DM^{-3}	1	2	5	10	20
NO INHIBITOR / MMOL MIN^{-1}	25	40	63	78	88
INHIBITOR P	12	22	45	66	83
INHIBITOR Q	12	20	32	39	44

a CALCULATE the percentage reduction in rate caused by Q at 20 mmol dm^{-3}. [2]

b IDENTIFY which inhibitor is competitive, giving evidence from the data. [2]

c EXPLAIN why adding more substrate does not overcome inhibitor Q. [2]

d A third inhibitor, R, stopped the reaction completely. Activity did not return when R was removed by dialysis. DEDUCE the type of inhibition, with a reason. [2]

e STATE one variable that must be kept constant. [1]

f EXPLAIN why initial rates were measured. [1]

Section D: Longer answer

6 MARKS

13 DISCUSS the use of enzyme inhibitors as medicines, using statins and penicillin as examples. [6]

[C1.1.15, C1.1.17]